Homework 11: Blood Spatter & Glass

Paper track. No home lab. Calculator with arcsin needed.

Part A: Angle of impact (2 pts each)

Compute the angle. `angle = arcsin(width ÷ length)`.

1. Width 3 mm, length 6 mm.
2. Width 4 mm, length 4 mm.
3. Width 2 mm, length 8 mm.
4. Width 6.9 mm, length 8.0 mm.
5. A stain gives an angle of impact of 90°. What did the stain look like, and how did the drop hit?

Part B: Spatter reasoning (2 pts each)

1. A stain is a long teardrop with the point aimed toward the window. Which way was the blood traveling?
2. Match the pattern to the cause: large drips and pools / a fine mist of tiny dots / a straight line of evenly spaced stains across the ceiling.
3. There is a clean, person-shaped blank area in the middle of a spatter pattern on the wall. What is this called, and what does it tell you?
4. You've drawn lines through the long axes of six stains and they all cross near one point. What have you found? What else do you need to find the area of *origin*?

Part C: Glass (2 pts each)

1. Which cracks form first — radial or concentric — and on which side of the force?
2. State the 3R rule.
3. A bullet hole in a window is narrow on the street side and wide on the room side. Was the shot fired from the street or from inside the room?
4. Two bullet holes, X and Y. The radial cracks from Y stop when they reach the cracks from X. Which shot was fired first?

5. Window pane A has a small clean hole with 2 short cracks. Pane B is shattered with a dozen long radial cracks. Which was hit by a high-speed projectile?

Answer Key

Part A:

1. arcsin(3/6) = arcsin(0.5) = **30°**.
2. arcsin(4/4) = arcsin(1) = **90°**.
3. arcsin(2/8) = arcsin(0.25) ≈ **14.5°**.
4. arcsin(6.9/8.0) = arcsin(0.8625) ≈ **59.6°**.
5. A **round** stain; the drop hit **straight down / perpendicular** to the surface.

Part B:

1. Toward the window (the tail points in the direction of travel).
2. Large drips/pools = low (passive) velocity; fine mist = high velocity (gunshot); a straight line across the ceiling = cast-off (flung from a swinging object).
3. A **void** — a person or object was in that spot, blocking the blood.
4. The **area of convergence** (the 2-D source location). You also need the **impact angles** of the stains to get the height / area of origin.

Part C:

1. Radial cracks form first, on the side **opposite** the force.
2. Radial cracks — the stress ridges meet the glass surface at a **R**ight angle on the **R**everse side (the side the force came from).
3. From the **street** — the cone-shaped hole widens in the direction the bullet traveled (narrow entry, wide exit), so narrow on the street side means it entered from the street.
4. **X** was fired first — Y's cracks stopped at X's cracks, so X's cracks were already there.
5. Pane A — a high-speed projectile punches a clean hole before the glass can flex and crack extensively.